

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

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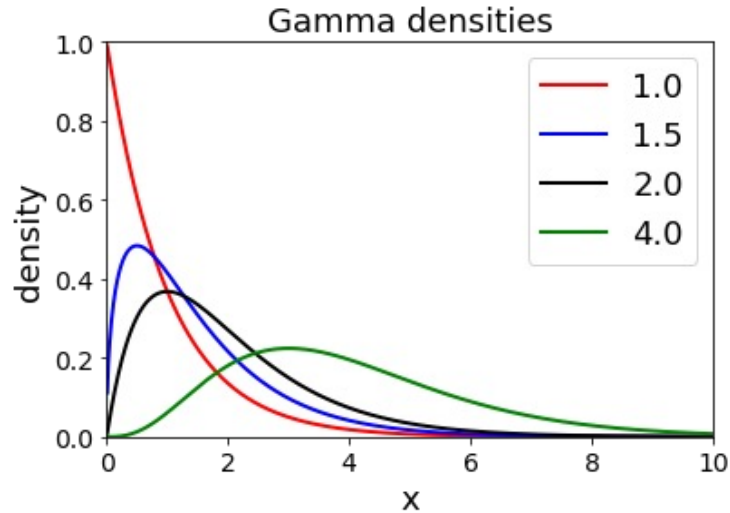


Poisson with a Gamma Prior

Recall Poisson(θ) distribution over $\mathbb{N}$: $p_{\theta}(x) = e^{-\theta} \theta^x / x!$.

Conjugate prior: gamma(α, β) distribution over $\mathbb{R}^+$:

$$\Pr(\theta) = \frac{\beta^{\alpha}}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta}.$$



- Mean: α/β
- Mode: $(\alpha - 1)/\beta$
- Variance: α/β^2

After seeing $x_1, \dots, x_n$, what is the posterior?

- Poisson distribution $p_\theta(x) = e^{-\theta} \theta^x / x!$
- Gamma(α, β) prior: $\Pr(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta}$